

3. Project Presentation

a. Context of the Project

The ONCF's role in Transportation Services

Moroccan National Railway Office (ONCF) is one of the key players in the development of transportation services in Morocco. plays a strategic role in the development of transportation services in Morocco. As the national railway operator, ONCF ensures the mobility of passengers and goods across the country while supporting economic growth, territorial connectivity, and sustainable transportation.

In order to guarantee operational reliability, safety, compliance, and service continuity, the Risk Management and Internal Audit Department plays a crucial role in mitigating risks and strengthening governance practices across the organization.

This department contributes to the organization by:

- Identifying and assessing operational, financial, and strategic risks
- Evaluating the effectiveness of internal control systems
- Ensuring compliance with regulations, standards, and internal procedures
- Supporting decision-making through audit recommendations and risk analysis
- Contributing to the continuous improvement of organizational performance and governance.

The methodology and problems currently faced by the workforce.

Although the department is strategically important, it still has a lot of reliance on traditional and manual working methods. Audit missions face some operational constraints which limit the efficiency and reach of the audit missions.

The current approach features:

- Hand picking and processing audit information.
- Data sources that are decentralized and fragmented.
- No centralized risk library and knowledge base.
- The lack of reporting, monitoring and analytical capacities.
- Collection and preliminary analysis of knowledge activities that require up to two weeks per mission

Such restrictions have several implications:

- Reduced annual audit coverage
- Significant time spent on repetitive administrative tasks
- Difficulty capitalizing on previous audit knowledge and expertise
- Limited visibility into risks and performance indicators
- Delays in producing reliable audit reports and recommendations

Digital Transformation Initiative at ONCF

To overcome these challenges and enhance the performance of the Internal Audit and Risk Management Department, ONCF has begun a digital transformation program that aims to modernize the audit processes. The goal of this effort is to leverage digital technologies, person-centric interfaces, data analytics and artificial intelligence to improve operational efficiency, expand mission coverage, and build mission risk management capability.

By leveraging intelligent and centralized solutions, the department aims to:

- Automate repetitive and time-consuming tasks
- Improve access to audit knowledge and historical data
- Accelerate risk analysis and reporting activities
- Support auditors with AI-assisted decision-making tools
- Increase the number and quality of audit missions conducted annually

Ultimately, the more risks the organization is capable of identifying, monitoring, and controlling, the more secure and resilient the company becomes.

b. Problematic

Main Problematic

How can ONCF digitalize and optimize its internal audit and risk management processes through AI-assisted and data-driven solutions while ensuring operational efficiency, standardization, security, and confidentiality of strategic information?

Research Questions

- How can the internal audit mission lifecycle be digitalized and optimized through intelligent workflows and centralized platforms?
- How can artificial intelligence assist auditors in time-consuming tasks such as knowledge collection, document analysis, and risk recommendation generation?

- How can AI technologies be integrated while ensuring the confidentiality, sovereignty, and security of ONCF's sensitive information?
- What application architecture best supports a hybrid human-AI collaboration model where automation assists auditors without replacing human expertise and decision-making?
- How can data analytics improve risk monitoring, reporting quality, and audit decision-making?
- How can a centralized audit knowledge base improve traceability, capitalization of expertise, and process standardization?

c. Proposed Solution

Aware of the operational and strategic challenges related to the current audit methodology, ONCF has decided to address this problematic through a digital transformation project conducted within the framework of this PFE internship.

Internship Subject

Design and Development of an AI-Assisted Internal Audit and Risk Management Platform for ONCF

The project aims to build a centralized and intelligent platform capable of supporting auditors throughout the audit lifecycle by combining process digitalization, data analytics, and artificial intelligence technologies.

Project Objectives

Primary Objectives

- Enhance auditor productivity through AI-assisted workflows
- Standardize audit processes across the organization
- Centralize audit-related data and documentation
- Improve audit quality and reporting accuracy

Strategic Impact

- Reduce audit preparation and execution time
- Improve compliance and traceability
- Enable data-driven audit decisions
- Strengthen internal control systems
- Increase audit mission coverage and operational efficiency
- Improve organizational resilience through better risk anticipation and monitoring